SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1741**

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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 10%

QUESTIONS:2 **Total Marks:20**

Descriptive Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behaviour and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

Code of Conduct:

I ____L Swetha_____ (192412678) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: L Swetha

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Understanding of Concepts	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions	5
Explanation	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation	5
Application	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis	4
Organization	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers	3
Accuracy	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps	2
Clarity	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand	1

ANSWERS

1) A)

Intelligent Agents in a Smart Home Automation System

1. Introduction

A smart home automation system is an intelligent system that automatically controls lighting, temperature, and security based on user behaviour, environmental conditions, and real-time sensor inputs. It uses sensors, controllers, and intelligent agents to make decisions with minimal human intervention.

For example, motion sensors detect the presence of people, temperature sensors monitor room conditions, and security sensors detect unauthorized entry. The system can also learn user preferences, such as preferred room temperature, lighting intensity, and sleeping schedules.

An intelligent agent is a system that perceives its environment through sensors and takes appropriate actions through actuators to achieve specific objectives.

The four major types of intelligent agents are:

1. Simple reflex agents
2. Model-based reflex agents
3. Goal-based agents
4. Utility-based agents

2. Types of Intelligent Agents and Their Functioning

A. Simple Reflex Agent

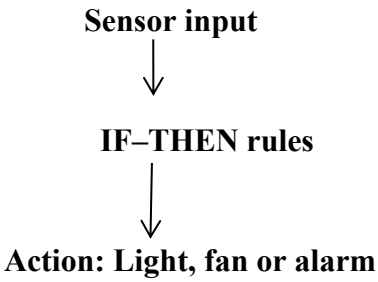
A simple reflex agent selects actions based only on the current percept (input) using predefined condition-action rules. It does not remember past events or learn user preferences.

Working principle:

The agent receives current sensor information and applies an IF–THEN rule to determine the action.

Examples:

- IF motion is detected, THEN switch ON the lights.
- IF the room temperature exceeds 28°C, THEN switch ON the fan.
- IF a door sensor detects unauthorized opening, THEN activate the alarm.



Advantages:

1. Simple to design and implement.
2. Provides fast responses to real-time inputs.
3. Requires low computational resources.

Limitations:

1. Cannot remember previous states or events.
2. Cannot learn user behaviour.
3. May make unsuitable decisions when sensor inputs are incomplete or conditions change.

Application: Suitable for basic automatic lighting, temperature thresholds, and immediate security alarms.

B. Model-Based Reflex Agent

A model-based reflex agent maintains an internal state or representation of the environment. It uses current sensor inputs and information from previous states to make better decisions.

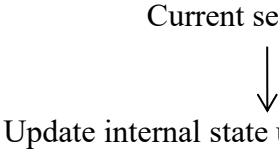
Unlike a simple reflex agent, it can consider what happened earlier and estimate the current condition of the home.

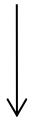
Working principle:

The agent updates its internal model using current sensor readings and previous information, then selects an action based on that model.

Examples:

1. It remembers that a person entered the living room and keeps the lights ON until the room is likely empty.
2. It considers the current temperature and the previous temperature trend before controlling the air conditioner.
3. It uses door status, motion history, and the current security mode to identify suspicious activity.





Apply rules and select action

Advantages:

- Maintains information about previous events.
- Handles situations where sensors provide incomplete information.
- Improves lighting, temperature, and security control.

Limitations:

- Requires additional memory and processing.
- Its performance depends on the accuracy of the internal model.
- Does not necessarily learn preferences or compare different possible outcomes.

Application: Suitable for occupancy tracking, temperature trend monitoring, and security systems that consider previous events.

C. Goal-Based Agent

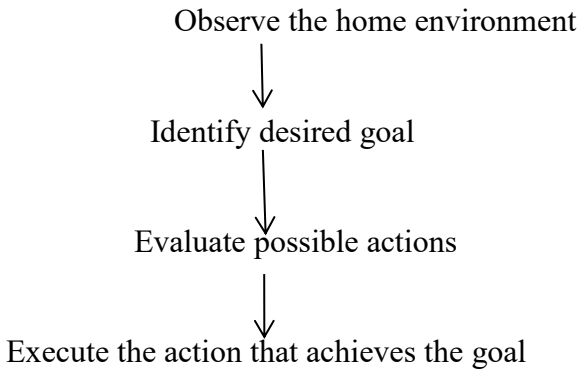
A goal-based agent selects actions by considering a specific goal or desired outcome. It evaluates possible actions and chooses those that help achieve the goal.In a smart home, goals may include maintaining a comfortable room temperature, reducing electricity consumption, or ensuring that the house is secure.

Working principle:

The agent observes the environment, identifies the current state, compares it with the desired goal, and selects an action to reduce the difference.

Examples:

- Goal: Maintain the room temperature at 24°C. The agent operates the air conditioner when necessary.
- Goal: Save electricity. The agent switches OFF unnecessary lights when rooms are unoccupied.
- Goal: Secure the house. The agent locks doors and activates security monitoring when the occupants leave.



Advantages:

1. Makes decisions based on desired outcomes.
2. Can plan actions rather than only reacting to sensor inputs.
3. Supports flexible automation for different user requirements.

Limitations:

1. May require more computation to evaluate possible actions.
2. A goal alone does not specify how to balance competing needs.
3. It may choose an action that achieves a goal but consumes excessive energy.

Application: Suitable for temperature regulation, energy-saving schedules, and automated home-security routines.

D. Utility-Based Agent

A utility-based agent evaluates possible actions using a utility function, which assigns a value to each outcome based on how desirable it is. It selects the action that provides the highest expected utility.This agent is particularly useful when comfort, energy efficiency, and security must be considered simultaneously.

Working principle:

The agent collects sensor data, estimates the outcomes of possible actions, calculates their utility, and chooses the action with the best overall trade-off.

For example, the system may compare several options:

1. Keeping the air conditioner ON provides comfort but consumes more electricity.
2. Increasing the temperature setting slightly saves energy but may reduce comfort.
3. Switching OFF the air conditioner saves energy but may make the room uncomfortable.

The agent chooses the option that best balances comfort and energy consumption while respecting security requirements.

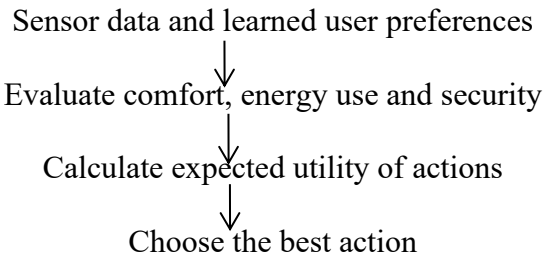
A simplified utility function can be represented as:

$$U(a)=w_cC(a)+w_eE(a)+w_sS(a)$$

Where:

- U(a) = utility of action a
- C(a) = comfort score
- E(a) = energy-efficiency score
- S(a) = security score
- (wc,we,ws) = weights representing the importance of each factor

In a practical system, these scores should be normalized, and safety-critical security requirements should be enforced as constraints rather than sacrificed for comfort or energy savings.



Advantages:

1. Balances multiple objectives.
2. Adapts decisions to changing environmental conditions.
3. Can incorporate user preferences learned over time.
4. Improves the trade-off between comfort and energy consumption.

Limitations:

1. Designing an appropriate utility function can be difficult.
2. Requires more computation and reliable sensor information.
3. Poorly selected weights may produce undesirable decisions.

Application: Suitable for intelligent HVAC control, personalized lighting, energy management, and coordinated home automation.

3. Comparison of the Four Intelligent Agents

Feature	Simple reflex	Model-based	Goal-based	Utility-based
Uses current sensor inputs	Yes	Yes	Yes	Yes
Remembers past states	No	Yes	Usually uses a state model	Usually uses a state model
Plans toward goals	No	Not necessarily	Yes	Yes, when needed
Compares multiple outcomes	No	Limited	Based on goal achievement	Based on utility
Learns user preferences	No	Not inherently	Not inherently	Can incorporate learning
Complexity	Low	Moderate	High	High
Smart home suitability	Basic automation	Context-aware control	Goal-oriented automation	Multi-objective optimization

Note: Goal-based and utility-based agents do not automatically learn. A separate learning mechanism can be added to update their models, goals, or utility parameters from user behaviour and feedback.

4. Analysis of the Agents in the Smart Home System

The four agents can perform different roles in the same home.

Smart home requirement	Most suitable agent	Reason

Turn lights ON when motion is detected	Simple reflex	Fast response using a direct rule
Track room occupancy and temperature trends	Model-based	Uses current and previous information
Maintain a preferred temperature	Goal-based	Works toward a desired temperature
Balance comfort and electricity consumption	Utility-based	Evaluates competing outcomes
Learn preferred lighting and temperature settings	Learning-enabled utility-based agent	Adjusts decisions using observed preferences
Respond immediately to security threats	Reflex rules with security constraints	Provides rapid, predictable protective actions

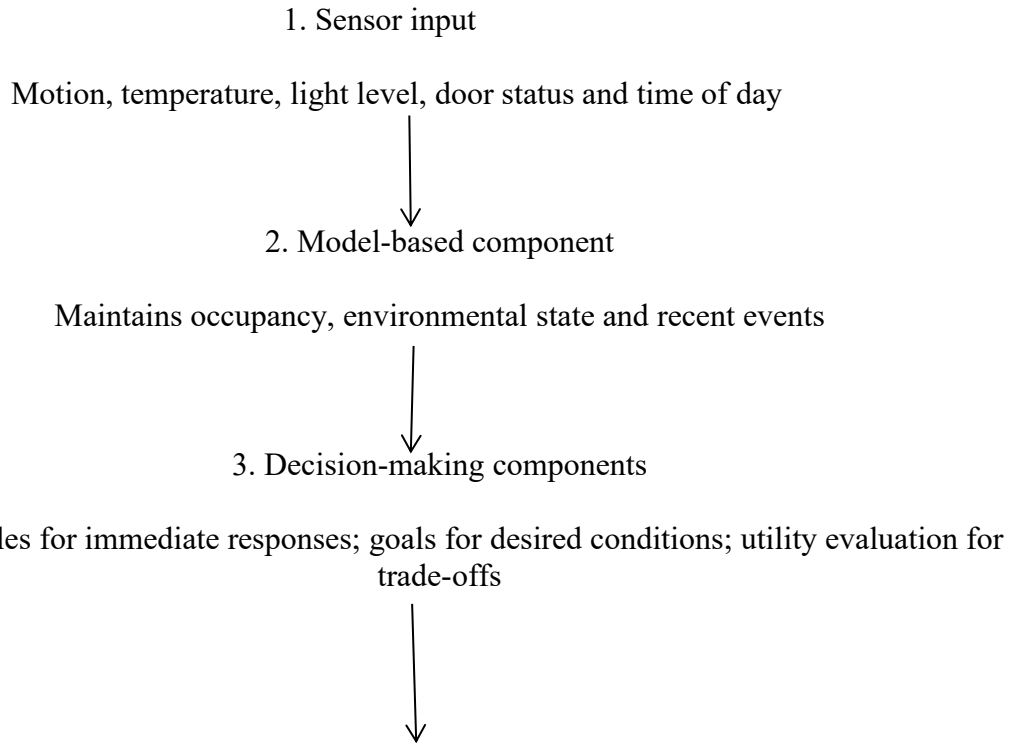
For example, when a person enters a room at night, the simple reflex component can switch on a light immediately. The model-based component can determine whether the room is occupied and whether the person is likely to be sleeping or moving between rooms. The goal-based component can maintain the desired temperature. Finally, the utility-based component can select an appropriate light intensity and air-conditioning setting to reduce energy use without compromising comfort.

5. Most Effective Approach: A Hybrid Intelligent Agent

The most effective solution is a hybrid intelligent agent combining all four approaches with a learning mechanism.

A single agent type cannot efficiently handle every smart home requirement. Simple reflex agents respond quickly, model-based agents understand context, goal-based agents pursue desired outcomes, and utility-based agents balance competing objectives.

A hybrid system can combine these strengths as follows:



4. Learning component

Updates user preferences from schedules, manual overrides and repeated behaviour

Why the hybrid approach is best

1. **Improved comfort:** The system learns preferred temperatures and lighting levels for different times of day and adjusts them automatically.

2. **Energy efficiency**: It switches off unnecessary lights and optimizes heating or cooling based on occupancy, schedules, and energy costs.

6. Example Scenario

Consider a smart home at 10:00 PM when a user enters the bedroom.

- ✓ Simple reflex agent: Detects motion and turns ON the bedroom light.
- ✓ Model-based agent: Uses the time, recent occupancy information, and room conditions to determine the context.

7. Conclusion

Intelligent agents play an important role in designing an efficient smart home automation system. Simple reflex agents are suitable for immediate actions, model-based agents support context-aware decisions, goal-based agents achieve desired environmental conditions, and utility-based agents optimize competing objectives.

2)A)

Uninformed Search Strategies for a Robot in an Unknown Maze

1. Introduction

A robot placed in an unknown maze must find the exit without prior knowledge of the layout. It explores possible paths, detects obstacles using sensors, and selects movements to reach the goal. Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) are two uninformed search strategies that can help solve this problem.

2. Uniform Cost Search (UCS)

UCS expands the path with the lowest total cost first.

Working:

- ➢ The robot starts from the initial position with cost zero.

- It explores available paths and calculates their total costs.
- It selects the path with the lowest cumulative cost.
- It continues until the exit is reached.

Advantages:

- Finds the least-cost path when movement costs are non-negative and standard conditions hold.
- Suitable when different paths require different amounts of energy or time.

Disadvantages:

- Requires high memory to store possible paths.
- Can take considerable time to explore many routes.

Complexity: Time and space can be exponential in the worst case.

3. Iterative Deepening Search (IDS)

IDS performs depth-limited searches repeatedly, increasing the depth limit each time until the exit is found.

Working:

- Starts searching at depth zero.
- Increases the depth limit gradually.
- Repeats the search from the starting position.
- Stops when the exit is found.

Advantages:

- Uses less memory than UCS.
- Finds the shallowest solution when the branching factor is finite.

Disadvantages:

- Repeats exploration of previously visited nodes.
- Does not guarantee the least-cost path when movement costs differ.

Complexity: Time $O(b^d)$; space $O(bd)$, where b is the branching factor and d is the depth of the shallowest goal.

4. Types of Intelligent Agents

- Simple Reflex Agent: Acts using current sensor readings but may repeat paths.
- Model-Based Agent: Remembers explored paths and detected obstacles.
- Goal-Based Agent: Selects actions to reach the exit.
- Utility-Based Agent: Chooses routes based on cost, safety, and energy consumption.
- Agent design affects how the robot stores information, handles uncertainty, and makes decisions.

5. Best Combination and Justification

A Model-Based Goal-Based Agent combined with UCS is suitable when movement costs differ.

- The model-based agent stores information about the explored maze.
- The goal-based agent plans actions toward the exit.
- UCS identifies the least-cost path.
- The robot can update its map and re-plan when it discovers new obstacles.
- If all movements have equal costs and memory is limited, IDS is a suitable alternative.

6. Conclusion

UCS is effective for finding the least-cost path but requires more memory. IDS is memory-efficient but repeats some exploration. A model-based goal-based agent combined with the appropriate search strategy helps the robot explore an unknown maze and reach the exit efficiently.